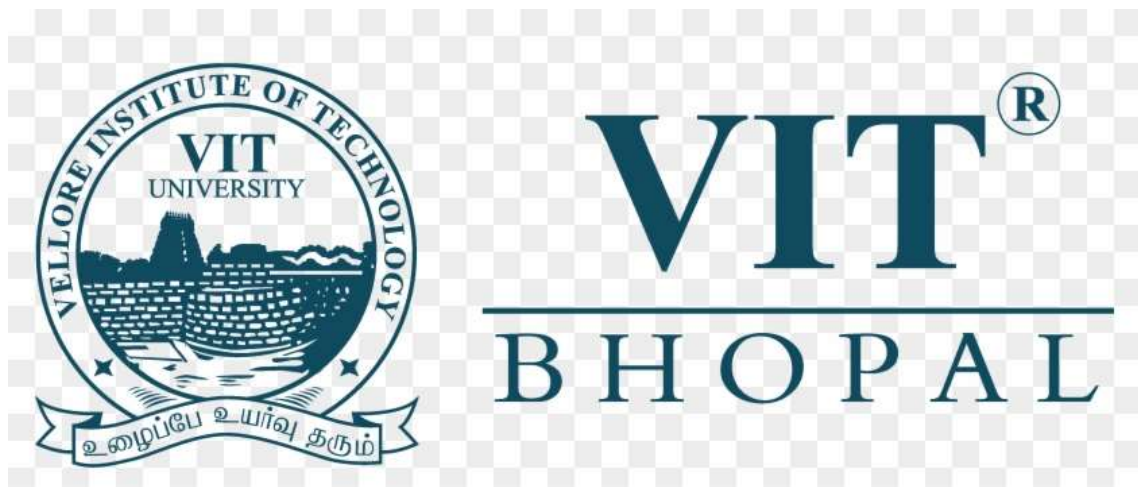


VITYARTHI PROJECT



ATM SIMULATION

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PROBLEM STATEMENT

- Develop a simple and user-friendly system to simulate ATM operations.
- Provide secure PIN-based authentication for accessing services.
- Allow users to check balance, deposit money, withdraw cash, and use Fast Cash.
- Maintain transaction records and display them through a Mini Statement.
- Validate user inputs and prevent invalid transactions.
- Provide an option to change the existing PIN securely.
- Reduce complexity by providing all major ATM operations through a menu-driven interface.

INTRODUCTION

The ATM Management System is a Python-based console application designed to simulate basic ATM operations. It provides users with a simple and interactive menu to perform banking activities such as PIN authentication, balance inquiry, cash withdrawal, money deposit, Fast Cash, Mini Statement, and PIN change.

The project demonstrates the practical use of Python programming concepts such as functions, loops, conditional statements, lists, strings, input validation, and global variables. It also helps in understanding how different functions can work together to create a complete menu-driven application.

DESIGN, DECISION AND RAIONALE

- Console-Based Interface: Chosen for simplicity and easy interaction without requiring a graphical interface.
- Python: Selected because it provides simple syntax and supports rapid development.
- Menu-Driven Design: A numbered menu allows users to easily select different operations.
- List for Transactions: A list is used to store deposit and withdrawal records during program execution.
- PIN Authentication: PIN verification is implemented to restrict access to ATM operations.
- Temporary Data Storage: Variables and lists are used instead of a database because the project is designed for learning and demonstration.
- Loop-Based Operation: A "while" loop keeps the ATM menu active until the user chooses to exit.

SYSTEM ARCHITECTURE AND DESIGN

Console Interface: Provides simple interaction without requiring a graphical interface.

Python: Enables rapid development with clear, concise syntax.

Modular Functions: Separates ATM operations to improve readability and maintainability.

Menu-Driven Design: Uses a numbered menu for straightforward operation selection.

Transaction List: Stores deposit and withdrawal records during program execution.

Input Validation: Prevents invalid amounts and menu selections.

PIN Authentication: Restricts access to authorized users.

Temporary Storage: Uses variables and lists instead of a database for learning and demonstration.

Loop-Based Operation: Keeps the menu active until the user exits.

FUNCTIONAL REQUIREMENTS

Functional Requirements

- PIN authentication
- Balance inquiry
- Cash withdrawal
- Money deposit
- Fast Cash
- Mini Statement
- PIN change
- Transaction tracking
- Input validation
- ATM menu navigation
- Secure exit

DESIGN DIAGRAMS

1. Start

2. Enter PIN

3. Check PIN

- Incorrect → Display error → End

- Correct → Continue

4. Display ATM Menu

5. Select Operation

6. Check Balance → Display Balance → Back to Menu

7. Withdraw Money → Enter Amount → Validate Amount & Balance → Update Balance → Back to Menu

8. Deposit Money → Enter Amount → Update Balance → Back to Menu

9. Fast Cash → Select Amount → Deduct Amount → Back to Menu

10. Mini Statement → Display Transactions → Back to Menu

11. Change PIN → Verify Old PIN → Enter New PIN → Update PIN → Back to Menu

12. Exit → Display Thank You → End

IMPLEMENTATION

- Developed the ATM application using Python 3.
- Implemented separate functions for each ATM operation.
- Used PIN authentication to control access to the system.
- Implemented a menu-driven interface using a "while" loop.
- Used conditional statements to process different user choices.
- Stored transaction details in a Python list.
- Added validation for withdrawal, deposit, and PIN inputs.
- Updated the account balance after successful transactions.
- Implemented Fast Cash with predefined withdrawal amounts.
- Added Mini Statement and PIN Change functionality.

OUTPUT

```
WELCOME TO VIT ATM :)
Enter your SECRET PIN: 7683
SECRET PIN is correct.

----- ATM.MENU -----
1. Check Balance
2. Withdraw Money
3. Deposit Money
4. Fast Cash
5. Mini Statement
6. Change PIN
7. Exit
Enter your choice: 1
Your current balance in Rs is . 75991.75
```

```
----- ATM.MENU -----
1. Check Balance
2. Withdraw Money
3. Deposit Money
4. Fast Cash
5. Mini Statement
6. Change PIN
7. Exit
Enter your choice: 7
Thank you for using the ATM :).
Have a nice day
(.venv) PS C:\Users\harsh\OneDrive\Desktop\Python>
```

TESTING

- Functional Testing: Tested each ATM operation individually.
- PIN Testing: Verified correct and incorrect PIN inputs.
- Balance Testing: Checked whether the displayed balance updates correctly.
- Withdrawal Testing: Tested valid amounts, insufficient balance, and invalid amounts.
- Deposit Testing: Tested valid, zero, and negative deposit amounts.
- Fast Cash Testing: Tested all predefined cash options and invalid choices.
- Transaction Testing: Verified that deposits and withdrawals appear in the Mini Statement.
- PIN Change Testing: Tested correct and incorrect old PINs and new PIN validation.
- Menu Testing: Tested valid and invalid menu choices.
- Exit Testing: Verified that the program terminates correctly after selecting Exit.

CHALLENGES

- Managing balance updates after deposits and withdrawals.
- Validating user inputs and handling invalid amounts.
- Implementing PIN authentication and PIN change functionality.
- Maintaining transaction history using a list.
- Designing a continuous menu using loops.
- Handling different user choices correctly.
- Managing multiple functions and their interaction within the program

LEARNING AND TAKEAWAYS

- Gained practical experience in Python programming.
- Improved understanding of functions and modular programming.
- Learned to use loops and conditional statements effectively.
- Practiced input validation and error handling.
- Learned to manage data using variables and lists.
- Understood how to design a menu-driven application.
- Improved logical thinking and problem-solving skills.
- Gained experience in developing and testing a real-world application.

FUTURE ENHACEMENTS

- Database integration for permanent data storage.
- Support multiple user accounts.
- Add transaction date and time.
- Implement limited PIN attempts and account blocking.
- Add GUI for a better user experience.
- Improve security by encrypting sensitive information.
- Add receipt generation for transactions.
- Develop a web or mobile version of the application.

CONCLUSION

The ATM Management System successfully demonstrates the practical application of Python programming concepts in a real-world scenario. Developing this project helped us understand the importance of logical thinking, proper program structure, validation, and testing.

We are grateful for the opportunity to work on this project and special thanks to our Professor Dr. M.K. Jayanthi which enhanced our programming knowledge and practical skills. We also appreciate the guidance and support received throughout its development.

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